ROCSCIENCE

Case2_with_surcharge
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:45:25 am
Software Version: 9.041

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Slide2 Analysis Information

Case2_with_surcharge

Project Summary

File Name:	Case2_with_surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.37s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:45:25 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	7
Tolerance:	0.005
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Grid Search
Radius Increment:	10
Composite Surfaces:	Disabled
Reverse Curvature:	Invalid Surfaces
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	10
Orientation:	Vertical

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	20 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	2.028950
Center:	13.689, 25.558
Radius:	15.989
Left Slip Surface Endpoint:	10.001, 10.000
Right Slip Surface Endpoint:	27.499, 17.500
Resisting Moment:	12232.6 kN-m
Driving Moment:	6029.01 kN-m
Total Slice Area:	49.1077 m2
Surface Horizontal Width:	17.4985 m
Surface Average Height:	2.80639 m

Method: spencer

FS	2.024780
Center:	13.689, 25.558
Radius:	15.989
Left Slip Surface Endpoint:	10.001, 10.000
Right Slip Surface Endpoint:	27.499, 17.500
Resisting Moment:	12207.4 kN-m
Driving Moment:	6029.01 kN-m
Resisting Horizontal Force:	667.895 kN
Driving Horizontal Force:	329.861 kN
Total Slice Area:	49.1077 m2
Surface Horizontal Width:	17.4985 m
Surface Average Height:	2.80639 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	1
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	1
Number of Invalid Surfaces:	0

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.02895

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.49979	40.9192	-8.80058	Material 1	20	20	13.1591	26.6991	18.4056	0	18.4056	16.3683	16.3683
2	2.49979	112.843	0.219983	Material 1	20	20	17.9427	36.4049	45.0723	0	45.0723	45.1412	45.1412
3	2.49979	164.92	9.24617	Material 1	20	20	21.0769	42.764	62.5437	0	62.5437	65.9748	65.9748
4	2.49979	196.306	18.5177	Material 1	20	20	22.5879	45.8297	70.9664	0	70.9664	78.5319	78.5319
5	2.49979	204.159	28.3422	Material 1	20	20	22.3466	45.3401	69.6213	0	69.6213	81.6749	81.6749
6	2.49979	181.923	39.2298	Material 1	20	20	19.9862	40.551	56.4634	0	56.4634	72.7811	72.7811
7	2.49979	81.0836	52.3794	Material 1	20	20	14.1718	28.7539	24.051	0	24.051	42.4398	42.4398

Global Minimum Query (spencer) - Safety Factor: 2.02478

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.49979	40.9192	-8.80058	Material 1	20	20	14.4861	29.3312	25.6373	0	25.6373	23.3946	23.3946
2	2.49979	112.843	0.219983	Material 1	20	20	19.2865	39.051	52.3421	0	52.3421	52.4161	52.4161
3	2.49979	164.92	9.24617	Material 1	20	20	21.8532	44.2479	66.6208	0	66.6208	70.1783	70.1783
4	2.49979	196.306	18.5177	Material 1	20	20	22.5657	45.6906	70.5844	0	70.5844	78.1425	78.1425
5	2.49979	204.159	28.3422	Material 1	20	20	21.552	43.6381	64.9449	0	64.9449	76.57	76.57
6	2.49979	181.923	39.2298	Material 1	20	20	18.7368	37.9379	49.284	0	49.284	64.5816	64.5816
7	2.49979	81.0836	52.3794	Material 1	20	20	13.4752	27.2842	20.0133	0	20.0133	37.4981	37.4981

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.02895

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	10.0005	10.0003	0	0	0
2	12.5003	9.61325	40.0062	0	0
3	15.0001	9.62285	84.4104	0	0
4	17.4999	10.0298	111.627	0	0
5	19.9997	10.8671	108.654	0	0
6	22.4995	12.2154	70.6197	0	0
7	24.9992	14.2564	5.32403	0	0
8	27.499	17.5	0	0	0

Global Minimum Query (spencer) - Safety Factor: 2.02478

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	10.0005	10.0003	0	0	0
2	12.5003	9.61325	46.2093	17.5578	20.8049
3	15.0001	9.62285	94.0191	35.7238	20.8049
4	17.4999	10.0298	121.65	46.2225	20.8049
5	19.9997	10.8671	119.078	45.2452	20.8049
6	22.4995	12.2154	85.495	32.485	20.805
7	24.9992	14.2564	31.8442	12.0996	20.8049
8	27.499	17.5	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 0 35, 0 35, 17.5 25, 17.5 10, 10 0, 10

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Distributed Load	25, 17.5 35, 17.5	Constant DistributionOrientation: VerticalMagnitude: 10 kN/m2Creates Excess Pore Pressure: No